Week 4 Notes & Quiz

Notes for all three Week 4 lectures, made only from the 86 slides in *WEEK 4.pdf*. Each yellow tag such as S38 is the slide (PDF page) the point comes from. The 40-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L18	**Trends in Logistics**	S1–S29
L19	**Trends in Logistics (contd.)**	S30–S65
L20	**Logistics vs SCM**	S66–S86
+	**Cheat sheet · Slide quirks · Quiz (40 Qs)**	—

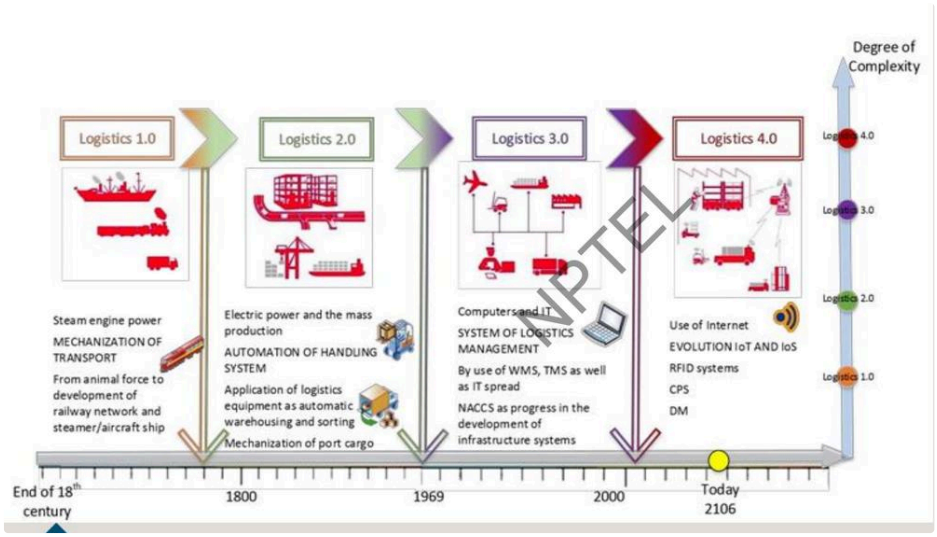
LECTURE 18 S1–S29

Trends in Logistics

Covers: history, categories, stakeholders, trends and technologies, cold chain, blockchain, big data S2

History of logistics

- Traced back to **military** contexts; computing advances in the **1960s and 1970s** brought mathematical modelling into logistics. S3
- Logistics became functional departments; third-party transport and LSPs professionalized; the idea of the **"Industrialization of Services"** took root. S3
- **UPS, FedEx and Walmart** pioneered the "Industrialization of Logistics Services". Then came E-Logistics, and now RFID and the **Internet of Things**. S4



Original slide S12 (Galindo, 2016). **Logistics 1.0**: steam power, mechanization of transport (railways, steamships). **2.0**: electric power, mass production, automated handling, automatic warehousing, mechanized port cargo. **3.0**: computers and IT, WMS and TMS, NACCS. **4.0**: internet, IoT and IoS, RFID, CPS. Timeline marks: end of 18th century, 1800, 1969, 2000.

Categories of modern logistics (Behrends, 2016) 7 items

Retail	Chain retailing and small independent stores, each with its own supply and delivery pattern. S7
Consumer shopping trips	Trips to shops and carrying goods home; big share of freight energy use. S7
Courier, Express, Parcel (CEP)	Global networks with cross-dock terminals and urban delivery tours. S7
HoReCa	**Hotel, Restaurants, Catering**: frequent Just-In-Time deliveries. S8
Construction	Material-intensive; irregular but timely supplies to sites. S8
Waste management	Specialized vehicles and pick-up tours to recycling and disposal sites. S9
Industrial and terminal haulage	Part and unit loads from manufacturing districts and terminals. S9

Stakeholders of modern logistics (Behrends, 2016) 3 groups

- **Shippers and receivers**: decide shipment size, frequency, lead time and precision, so they **determine transport demand**. S10
- **Carriers**: provide transport services, own-account or third-party; aim for high resource utilization. S11
- **Public authorities**: land-use and traffic planning, roads and rail; balance economic activity with residents' quality of life. S11

Trends and technologies

- Trends centre on **automation, sustainability and technology integration**. Future trends: elastic logistics, green logistics, last-mile delivery, autonomous vehicles, on-demand warehousing. S13

- Technologies: automation, blockchain, IoT, advanced analytics, cloud computing, robotics, AI, augmented reality, digital twins, advanced geolocation. S13

Elastic logistics	Rapidly **shrink or expand** logistics activities and infrastructure in line with demand, using data. S14
Blockchain	Finds where standards are breached or goods damaged; **smart contracts** cut fraud and admin errors. S14
IoT	Automates warehouse operations, optimizes space. S14
Cloud computing	On-demand access to logistics software, storage and processing. S14
Logistics robots	Store and move goods inside warehouses: **intralogistics**. S15
Augmented reality	AR warehousing apps find the shortest routes to items. S15
Digital twins	**Scenario testing** (disruptions, black swan events) without touching real operations. S15
Geolocation	GPS, BLE, Wi-Fi, cellular, IoT trackers. S15

Drone system for last-mile delivery: depots, trucks and drones serving customers (Eskandaripour & Boldsaikhan, 2023). S17

Modern concept 1: cold chain logistics

- Managing and transporting **temperature-sensitive** products (perishables, pharmaceuticals) in a controlled environment, from refrigeration to freezing. S18
- Key components: **temperature control**, packaging (insulated containers, refrigerated trucks), monitoring and tracking, risk management (temperature excursions), regulatory compliance (**FDA, WHO**). S18–S19

Modern concept 2: blockchain

- A **decentralized, distributed ledger** that records transactions securely and immutably **without intermediaries**. S20
- Elements: transparency and traceability, enhanced visibility, improved security (cryptographic encryption, decentralized consensus), streamlined processes (**smart contracts**: self-executing contracts with predefined rules), reduced costs and delays. S21–S22

Modern concept 3: big data

Roles: predictive analytics · route optimization · warehouse management · demand forecasting · supply chain visibility · risk management. S23–S25

LECTURE 19 S30–S65

Trends in Logistics (contd.)

Factors fostering innovation 11 items

Supply chain agility	Self-managed onboarding: start trading with a new client in hours instead of waiting weeks in a managed-service queue. S33
Global labour shortages	Push toward automation; driverless truck pairs where the second truck copies a human-driven lead truck. S34
Automation complexity	Integrate WMS/ERP with e-commerce platforms to stop manual, error-prone data hunting. S35
Real-time data demand	The **"Amazon effect"** (2-day and same-day shipping); visibility platforms such as **project44** and **FourKites**. S36
Transparency and collaboration	**EDI + APIs** together: API gives deeper insight, EDI starts and orchestrates processes. S37
Less Than Truckload (LTL)	Several small orders from different companies on one carrier. Benefits: pay only for space used, suits e-commerce, small-business friendly, fewer trips and emissions. S38
Digital transformation · APIs	APIs do **real-time processing instead of batch processing**. S39
Demand forecasting	Estimate future demand from sales history, seasonality, economy, pricing, competition. S40
Digital freight marketplaces	Online networks where shippers compare carriers' times, prices and services in one place. S40
Ecosystem integration	One all-encompassing integration platform instead of many tools: fewer errors, cheaper, real-time data flow. S41

Trends in India

TREND	INDIAN EXAMPLE
Digital transformation	IoT fleet management: vehicle performance, fuel, driver behaviour. S42
Last-mile innovations	**Flipkart** piloted drone deliveries. S43
Sustainable practices	**Amazon India**: 100% electric delivery fleet by **2030**. S44
Warehousing automation	**DHL eCommerce India** automated sortation centre in Delhi: up to **6,000 shipments per hour**. S45
Predictive analytics	**Delhivery** optimizes delivery routes. S47
Collaboration and integration	**BlackBuck** digital freight platform connects shippers and truckers. S48
E-commerce logistics focus	**Ekart**, Flipkart's logistics arm. S49

Challenges for modern logistics 5 items

- **Supply chain disruptions**: 2023 Canadian wildfires, California floods, Russia–Ukraine and Israel–Hamas wars, strikes, low water at the Panama Canal. S51
- **Trading partners failing commitments**: **80%** of companies said partners failed; 27% had 11+ partners fail; 29% plan new partnerships. S52
- **Rising operational costs**: top hurdle is rising cost of goods and labour; 32% will invest in supply chain tech; 43% say ecosystem integration cuts costs, 33% say it raises revenue. S53
- **Sustainability targets**: **45%** call sustainability a critical objective; 32% say ecosystem integration removes data silos. S54
- **Labour shortages**. S50

Performance measures for modern logistics 7 items

Efficiency · cost-effectiveness · customer satisfaction · environmental impact · flexibility and agility · innovation adoption · supply chain visibility. S55–S56

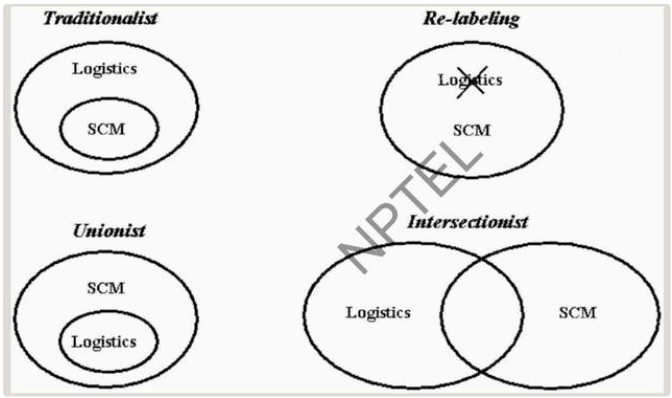
Case studies

- **DHL** invested **$350 million** in digitization and built **MySupplyChain**, an end-to-end visibility solution using predictive analytics. S57
- **Amazon**: **anticipatory shipping** predicts when, where and which items customers will buy, so popular items ship from a nearby hub. S58
- **Swiggy**: **hyperlocal** delivery networks of local restaurants and delivery partners. S59
- **Flipkart**: partners with **local logistics providers** for coverage in remote areas. S60

LECTURE 20 S66–S86

Logistics vs SCM

- The terms are often used interchangeably; some call SCM the "new" logistics. What the US calls SCM is more commonly **logistics management in Europe**. S68
- Inbound Logistics readers: "There isn't a difference today" (Wayne Johnson, American Gypsum); "SCM incorporates logistics, and logistics is a number of sub-processes within SCM" (Michael Kirby). S69



Original slide S70, four perspectives. **Traditionalist**: SCM is a part of logistics. **Re-labeling**: logistics renamed as SCM. **Unionist**: logistics is a part of SCM. **Intersectionist**: the two overlap.

> LOGISTICS (CSCMP) · EXACT WORDS
>
> *"part of the supply chain process that plans, implements and controls the efficient, effective forward and reverses flow and storage of goods, services and related information between the point of origin and the point of consumption in order to meet customer's requirements."* S73

- **SCM** (Michigan State, *Supply Chain Logistics Management*): collaboration between firms to connect suppliers, customers and partners; SCM decisions are **strategic** and set "the operational framework within which logistics is performed". S72
- Logistics covers demand planning, inventory, transportation, fleet management, material handling, order fulfilment. **Inbound**: acquisition, handling, transport and storage of materials. **Outbound**: collection, maintenance, distribution to the end user. S74
- Objective of logistics: desired product at the right **time, quality, place and price**. S74
- What they share: both deal with goods, services or information; both raise customer satisfaction; both support competitive success; same flow from supplier to manufacturer to wholesaler to retailer. S75

	SUPPLY CHAIN MANAGEMENT	LOGISTICS
Main focus	**Competitive advantage**	**Meeting customer requirements**
Age of term	Relatively new; **Keith Oliver** coined it in the early **1980s**	Old; from the military; **Alexander the Great** (b. 356 BC) a logistics master
Scope	Production and inventory planning, labour, materials, facilities, manufacturing, delivery	Movement, storage and flow of goods, services and information
Relation	Links business processes within and across companies	An **activity within** the supply chain

Differences from S76–S77.

Logistical components of the supply chain 8 items

Information · storage · warehousing · material handling · packaging · **unitization** (making items efficient to arrange, move and store) · inventory control · transportation. S79–S80

SC STAGE	LOGISTICS COMPONENTS USED S81
Planning	Information, inventory control
Sourcing	Information, material handling, transportation
Manufacturing	Information, material handling, storage, warehousing, packaging, unitization, transportation
Delivering	Information, warehousing, inventory management, materials handling, packaging, transportation
Returning	Information, packaging/unitization, material handling, transportation, warehousing

Case: GEODIS and an online retailer

- Problems: big **seasonal swings**, next-day or two-day delivery promise, accurate inventory. S82
- Solutions: KPI monitoring, demand forecasting, scalable **pop-up warehouses**, shipping from warehouses nearest transport hubs. S82–S83
- Result: a multi-country network and a 91% fulfilment rate for European customers. S83

BEFORE THE EXAM ALL LECTURES

Cheat sheet

NUMBER / NAME	MEANING	SLIDE
1960s–70s	Computing brings maths modelling to logistics	S3
7 · 3	Behrends: logistics categories · stakeholder groups	S6–S11
2030	Amazon India 100% EV delivery fleet	S44
6,000/h	DHL eCommerce Delhi sortation centre	S45
80% · 45%	Partners failed commitments · sustainability critical	S52 S54
$350 m	DHL digitization, MySupplyChain	S57
1980s	Keith Oliver coins "supply chain management"	S77
356 BC	Alexander the Great, logistics master	S77
91%	GEODIS fulfilment rate in Europe	S83

Company → concept	Flipkart → drones, local partners · Amazon → anticipatory shipping · Swiggy → hyperlocal · BlackBuck → freight platform · Ekart → e-commerce logistics · Delhivery → predictive routing
Logistics 1.0 → 4.0	Steam/mechanized transport → electricity/automation → IT, WMS/TMS → internet, IoT, RFID, CPS
4 perspectives	Traditionalist, re-labeling, unionist, intersectionist

MEMORY HOOK (MY OWN, NOT FROM THE SLIDES)

Unionist = **U**nder SCM: logistics sits *inside* the SCM circle. Traditionalist is the reverse.

READ ONCE SOURCE CHECK

Slide quirks to watch

- The Logistics 1.0–4.0 timeline labels today as "2106"; it should be 2016 (Galindo, 2016). S12
- The CSCMP definition is printed with "forward and reverses flow" (a typo for reverse). S73
- "Labor shortages" appears in the challenges diagram (S50) but gets no slide of its own; it is covered as a factor on S34.
- Lecture 18 and 19 show the same topic list (S2, S31), though L18 covers history, categories and technologies and L19 covers factors, India, challenges and cases.
- Video or link-only slides: S5, S16, S46. The title slide for Lecture 20 reads "Lecture 20 – : Logistics vs SCM".

SELF-TEST 40 QUESTIONS

Week 4 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 18

According to Lecture 18, logistics can be traced back to which context?

A Military
B Retail
C Banking
D Agriculture

Q2 LECTURE 18

Which companies are credited with pioneering the 'Industrialization of Logistics Services'?

A Delhivery, Ekart, BlackBuck
B DHL, Maersk, Toyota
C UPS, FedEx, Walmart
D Amazon, Flipkart, Alibaba

Q3 LECTURE 18

In Galindo's timeline, which stage is associated with the internet, IoT and IoS, RFID and CPS?

A Logistics 2.0
B Logistics 1.0
C Logistics 3.0
D Logistics 4.0

Q4 LECTURE 18

Which Behrends category involves food and beverage deliveries to hotels, restaurants and caterers, often on a JIT basis?

A CEP
B HoReCa
C Retail
D Terminal haulage

Q5 LECTURE 18

CEP services in modern logistics stand for:

A Container, Equipment, Pallet
B Courier, Express and Parcel
C Central Express Packaging
D Cargo, Export, Port

Q6 LECTURE 18

Which stakeholder's logistics decisions determine transport demand (shipment size, frequency, lead time)?

A Carriers
B Shippers and receivers
C Public authorities
D Customs agents

Q7 LECTURE 18

'The ability of a business to rapidly reduce or expand logistics activities and infrastructure in line with demand' is:

A Lean logistics
B Elastic logistics
C Agile logistics
D Reverse logistics

Q8 LECTURE 18

Using robots to store and move goods inside warehouses is called:

A cold chain
B interlogistics
C reverse logistics
D intralogistics

Q9 LECTURE 18

Which technology is used for scenario testing of disruptions and black swan events without affecting real operations?

A Blockchain
B Digital twins
C Cloud computing
D RFID

Q10 LECTURE 18

Cold chain logistics regulatory compliance mentions which authorities?

A FDA and WHO
B WTO and IMF
C ISO and IEC
D EPA and OSHA

Q11 LECTURE 18

Blockchain is described as a:

A type of RFID tag
B warehouse robot
C decentralized and distributed ledger that records transactions without intermediaries
D centralized database run by a bank

Q12 LECTURE 18

Self-executing contracts with predefined rules that are enforced automatically are called:

A EDI messages
B Incoterms
C smart contracts
D bills of lading

Q13 LECTURE 18

Which is NOT listed as a role of big data in logistics?

A Route optimization
B Demand forecasting
C Risk management
D Replacing all human drivers

Q14 LECTURE 18

The drone system for last-mile delivery shown on the slide comes from:

A Behrends (2016)
B Galindo (2016)
C Eskandaripour & Boldsaikhan (2023)
D Klaus (2009)

Q15 LECTURE 19

Customers expecting constant updates on order location because of fast 2-day and same-day shipping is called the:

A Halo effect
B Network effect
C Amazon effect
D Bullwhip effect

Q16 LECTURE 19

Which supply chain visibility platforms are named on the slides?

A SAP and Oracle
B BlackBuck and Delhivery
C project44 and FourKites
D Shopify and Magento

Q17 LECTURE 19

In Less Than Truckload (LTL) shipping:

A shipments are never consolidated
B one customer books an entire truck
C multiple smaller orders from various companies share one carrier
D goods move only by rail

Q18 LECTURE 19

Why are logistics companies moving to API-based integrations?

A APIs remove the need for EDI entirely
B APIs are cheaper than paper
C APIs perform real-time processing rather than batch processing
D APIs use batch processing

Q19 LECTURE 19

Digital networks where shippers and carriers connect and compare prices, times and services are:

A hyperlocal networks
B digital freight marketplaces
C ecosystem integration
D elastic logistics

Q20 LECTURE 19

Using one all-encompassing integration platform instead of multiple tools is called:

A EDI modernization
B Ecosystem integration
C Cloud migration
D Self-managed onboarding

Q21 LECTURE 19

Amazon India has committed to converting its delivery fleet to 100% electric vehicles by:

A 2030
B 2050
C 2040
D 2025

Q22 LECTURE 19

DHL eCommerce India's automated sortation centre in Delhi can handle up to:

A 6,000 shipments per day
B 60,000 shipments per day
C 600 shipments per hour
D 6,000 shipments per hour

Q23 LECTURE 19

Which Indian company is cited for connecting shippers and truckers through a digital freight platform?

A Delhivery
B Swiggy
C BlackBuck
D Ekart

Q24 LECTURE 19

In the ecosystem integration report, what share of companies said trading partners failed to uphold business commitments?

A 80%
B 29%
C 27%
D 45%

Q25 LECTURE 19

What share of businesses said sustainability was a critical business objective?

A 43%
B 45%
C 80%
D 32%

Q26 LECTURE 19

DHL invested $350 million to build which end-to-end visibility solution?

A SmartTrucking
B MySupplyChain
C Resilience360
D GoGreen

Q27 LECTURE 19

Amazon's practice of predicting when, where and which items customers will buy is called:

A anticipatory shipping
B drop shipping
C hyperlocal delivery
D cross-docking

Q28 LECTURE 19

Swiggy revolutionized last-mile delivery mainly through:

A hyperlocal delivery networks
B drone delivery
C rail freight
D 5PL aggregation

Q29 LECTURE 20

What the US calls supply chain management is more commonly known in Europe as:

A procurement
B materials management
C logistics management
D operations management

Q30 LECTURE 20

In the 'unionist' perspective:

A logistics was simply renamed SCM
B SCM is part of logistics
C logistics and SCM partially overlap
D logistics is part of SCM

Q31 LECTURE 20

The perspective in which logistics and SCM partially overlap is called:

A Unionist
B Re-labeling
C Intersectionist
D Traditionalist

Q32 LECTURE 20

Which body's definition calls logistics 'part of the supply chain process that plans, implements and controls...'?

A World Trade Organization
B Council of Supply Chain Management Professionals
C Michigan State University
D ISO

Q33 LECTURE 20

Material acquisition, handling, transport and storage of commodities are part of:

A inbound logistics
B reverse logistics
C outbound logistics
D green logistics

Q34 LECTURE 20

The main focus of supply chain management, as opposed to logistics, is:

A fleet management
B competitive advantage
C warehousing
D meeting customer requirements

Q35 LECTURE 20

Who is credited with coining the term 'supply chain management' in the early 1980s?

A Taiichi Ohno
B Michael Porter
C Peter Drucker
D Keith Oliver

Q36 LECTURE 20

Which historical figure is called a logistics master on the slides?

A Napoleon
B Julius Caesar
C Genghis Khan
D Alexander the Great

Q37 LECTURE 20

Making items efficient to arrange, transport and store so handling equipment can move them without damage is:

A packaging
B unitization
C palletless loading
D consolidation

Q38 LECTURE 20

In the role of logistics within SCM, which SC stage uses only 'information and inventory control'?

A Planning
B Delivering
C Returning
D Sourcing

Q39 LECTURE 20

GEODIS handled the online retailer's peak-season surges using:

A scalable pop-up warehouses
B rail freight
C a 5PL aggregator
D drones

ANSWERS 40 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	S3: logistics traces back to military contexts; computing in the 1960s–70s brought mathematical modelling. S3
Q2	C	S4. S4
Q3	D	1.0 steam, 2.0 electricity and automation, 3.0 computers/WMS/TMS, 4.0 internet/IoT/RFID/CPS. S12
Q4	B	HoReCa = Hotel, Restaurants and Catering. S8
Q5	B	S7: Couriers, Express and Parcel services using cross-dock terminals. S7
Q6	B	S10: shippers and receivers. S10
Q7	B	S14: elastic logistics, using data-driven technology. S14
Q8	D	S15. S15
Q9	B	S15: digital twins. S15
Q10	A	S19: FDA and WHO. S19
Q11	C	S20. S20
Q12	C	S22: smart contracts. S22
Q13	D	Roles: predictive analytics, route optimization, warehouse management, demand forecasting, visibility, risk management. S23–S25
Q14	C	S17. S17
Q15	C	S36: the 'Amazon effect'. S36
Q16	C	S36 and S54. S36
Q17	C	S38: customers pay only for the space they use. S38
Q18	C	S39. S39
Q19	B	S40. S40
Q20	B	S41. S41
Q21	A	S44. S44
Q22	D	S45. S45
Q23	C	S48. S48
Q24	A	80%; 27% had 11 or more partners fail; 29% plan new partnerships. S52
Q25	B	S54: 45%. S54
Q26	B	S57. S57
Q27	A	S58. S58
Q28	A	S59. S59
Q29	C	S68. S68
Q30	D	Unionist: logistics inside SCM. Traditionalist: SCM inside logistics. Re-labeling: renamed. Intersectionist: overlap. S70
Q31	C	S70. S70
Q32	B	S73: CSCMP. S73
Q33	A	S74. S74
Q34	B	SCM → competitive advantage; logistics → meeting customer requirements. S76
Q35	D	S77. S77
Q36	D	Alexander the Great, born 356 BC. S77
Q37	B	S80: unitization. S80
Q38	A	S81. S81
Q39	A	S82: plus KPI monitoring, demand forecasting and shipping from warehouses near transport hubs. S82
Q40	D	S83: 91%. S83

Made from *WEEK 4.pdf* (86 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".